**P1** (a) $-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\psi(x,y)+V(x,y)\,\psi(x,y)=E\,\psi(x,y)$

$$V(x,y)=\begin{cases}0, & \text{if } 0\le x\le a \text{ and } 0\le y\le b\\ \infty, & \text{otherwise}\end{cases}$$

let $\psi(x,y)=X(x)\,Y(y)$ :

$$-\frac{\hbar^2}{2m}\left(Y\frac{d^2X}{dx^2}+X\frac{d^2Y}{dy^2}\right)=E\,X(x)\,Y(y) \quad \text{(inside the well)}$$

divide by $XY$ :

$$\underbrace{-\frac{\hbar^2}{2m}\frac{1}{X}\frac{d^2X}{dx^2}}_{E_x}\;\underbrace{-\frac{\hbar^2}{2m}\frac{1}{Y}\frac{d^2Y}{dy^2}}_{E_y}=E$$

$$\Rightarrow 2\ \text{S.E:}\begin{cases}-\frac{\hbar^2}{2m}\frac{d^2X}{dx^2}=E_x X & (0\le x\le a)\\ -\frac{\hbar^2}{2m}\frac{d^2Y}{dy^2}=E_y Y & (0\le y\le b)\end{cases}$$

$$E=E_x+E_y=\frac{\pi^2\hbar^2}{2ma^2}n_x^2+\frac{\pi^2\hbar^2}{2mb^2}n_y^2$$

$$\psi(x,y)=\underbrace{\sqrt{\frac{2}{a}}\sin\left(\frac{\pi n_x}{a}x\right)}_{X_{n_x}(x)}\;\underbrace{\sqrt{\frac{2}{b}}\sin\left(\frac{\pi n_y}{b}y\right)}_{Y_{n_y}(y)}$$

(b) let $a=\sqrt{2}\,b$ , $E=\frac{\pi^2\hbar^2}{2mb^2}\left(\frac{1}{2}n_x^2+n_y^2\right)$.

| $n_x$ | $n_y$ | $E\cdot 2mb^2/\pi^2\hbar^2$ | | |
|---|---|---|---|---|
| 1 | 1 | $3/2$ | ← ground state | non-degenerate |
| 2 | 1 | $3$ | ← 1st excited | |
| 1 | 2 | $\frac{1}{2}+4=9/2$ | ← 2nd excited | |
| 3 | 1 | $9/2+1=11/2$ | | |

(c) For $a=b$, $E = \frac{\pi^2\hbar^2}{2mb^2}(n_x^2+n_y^2)$

$\Rightarrow$ ground state $(n_x = n_y = 1)$ is non-degenerate

excited states $(n_x=2, n_y=1)$ and $(n_x=1, n_y=2)$ have the same energy $\Rightarrow$ doubly degenerate

(d) We need $(n_x=3, n_y=1)$ be the 2nd excited state

$$\Rightarrow \quad E(n_x=3, n_y=1) < E(n_x=1, n_y=2) \quad (1)$$

$$\text{and } E(3,1) > E(2,1) \quad (2)$$

From (1): $\frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2}9+\frac{1}{b^2}\right) < \frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2}+\frac{4}{b^2}\right)$

$$\frac{8}{a^2} < \frac{3}{b^2} \quad (1')$$

From (2): $\frac{\pi^2\hbar^2}{2m}\left(\frac{9}{a^2}+\frac{1}{b^2}\right) > \frac{\pi^2\hbar^2}{2m}\left(\frac{4}{a^2}+\frac{1}{b^2}\right)$

$$\frac{5}{a^2} > 0 \quad \text{(always fulfilled)}$$

$\Rightarrow$ from (1') $\boxed{\frac{a}{b} > \sqrt{\frac{8}{3}}}$

P2 $\Psi(x,0) = A\left(3 + \sqrt{\frac{8m\omega}{\hbar}}\, x\right) e^{-m\omega x^2/2\hbar}$

(a) From [2.59] $e^{-\frac{m\omega}{2\hbar}x^2} = \left(\frac{\pi\hbar}{m\omega}\right)^{1/4} \psi_0(x)$

[2.62] $\sqrt{\frac{2m\omega}{\hbar}}\, x\, e^{-m\omega x^2/2\hbar} = \left(\frac{\pi\hbar}{m\omega}\right)^{1/4} \psi_1(x)$

$$\Rightarrow \Psi(x,0) = A\left(\frac{\pi\hbar}{m\omega}\right)^{1/4}\left(3\psi_0 + 2\psi_1\right)$$

Normalize: $1 = |A|^2\left(\frac{\pi\hbar}{m\omega}\right)^{1/2} \underbrace{\int dx\, (3\psi_0^* + 2\psi_1^*)(3\psi_0 + 2\psi_1)}_{9\int dx\, |\psi_0|^2 + 4\int dx\, |\psi_1|^2 = 13}$

Because $\int dx\, \psi_0(x)\psi_1(x) = 0$ by orthogonality!

Thus $A = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{13}}$

(b) $\Psi(x,0) = \frac{3}{\sqrt{13}}\psi_0 + \frac{2}{\sqrt{13}}\psi_1$

$$\Psi(x,t) = \frac{3}{\sqrt{13}}\psi_0 e^{-iE_0 t/\hbar} + \frac{2}{\sqrt{13}}\psi_1 e^{-iE_1 t/\hbar} =$$

$$= e^{-i\omega t/2}\left(\frac{3}{\sqrt{13}}\psi_0 + \frac{2}{\sqrt{13}}\psi_1 e^{-i\omega t}\right).$$

(c) $$\langle x\rangle = \int dx\, \frac{e^{i\omega t/2}}{\sqrt{13}}\left(3\psi_0 + 2\psi_1 e^{i\omega t}\right) x\, \frac{e^{-i\omega t/2}}{\sqrt{13}}\left(3\psi_0 + 2\psi_1 e^{-i\omega t}\right)$$

$$= \frac{1}{13}\Big\{\underbrace{\int dx\; 9|\psi_0|^2 x}_{\emptyset} + \underbrace{\int dx\; 4|\psi_1|^2 x}_{\emptyset \text{ by oddness}} +$$

$$+ 6\int dx\left(\psi_1\psi_0\, x\, e^{i\omega t} + \psi_0\psi_1\, x\, e^{-i\omega t}\right)\Big\}.$$

$$\langle x \rangle = \frac{6}{13} \cdot 2\cos\omega t \int_{-\infty}^{\infty} dx \; x\, \psi_0(x)\, \psi_1(x) =$$

$$= \frac{12}{13}\cos\omega t \int_{-\infty}^{\infty} dx \; x \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \sqrt{\frac{2m\omega}{\hbar}}\; x\, e^{-\frac{m\omega x^2}{\hbar}} =$$

$$\left(x = \sqrt{\frac{\hbar}{m\omega}}\, y\right) \qquad = \frac{12}{13}\cos\omega t \int \sqrt{\frac{\hbar}{m\omega}}\, dy \left(\frac{2}{\pi}\right)^{1/2} \frac{m\omega}{\hbar} \frac{\hbar}{m\omega}\, y^2 e^{-y^2}$$

$$= \frac{12\cos\omega t}{13} \sqrt{\frac{2\hbar}{\pi m\omega}} \underbrace{\int_{-\infty}^{\infty} dy\, y^2 e^{-y^2}}_{2\cdot\sqrt{\pi}\,\frac{2!}{1!}\left(\frac{1}{2}\right)^3 \;=\; \frac{\sqrt{\pi}}{2}}$$

$$\langle x \rangle = \frac{12\cos\omega t}{13} \sqrt{\frac{\hbar}{2m\omega}}$$

$$\langle p \rangle = m\frac{d\langle x\rangle}{dt} = m\sqrt{\frac{\hbar}{2m\omega}}\; -\omega\, \frac{12\sin\omega t}{13} =$$

$$= -\sqrt{\frac{\hbar m\omega}{2}}\; \frac{12}{13}\sin\omega t$$

(d) Measurement of energy will give:

$$\begin{cases} E_0 = \frac{1}{2}\hbar\omega \\ \text{with } P_0 = \frac{9}{13} \\ E_1 = \frac{3}{2}\hbar\omega \\ \text{with } P_1 = \frac{4}{13}. \end{cases}$$

(e) $$\psi(x,T) = B\left(3 - 2\sqrt{\frac{2m\omega}{\hbar}}\, x\right) e^{-m\omega x^2/2\hbar} =$$

$$= e^{-i\omega T/2}\left(\frac{3}{\sqrt{13}}\psi_0 + \frac{2}{\sqrt{13}}\psi_1 e^{-i\omega T}\right)$$

$$\Rightarrow e^{-i\omega T} = -1 \Rightarrow \omega T = \pi \Rightarrow T = \pi/\omega.$$

P3 $\psi(t=0) = \frac{4}{5}\psi_{100} + \frac{3i}{5}\psi_{211}$

$\psi_{nlm}$ are eigenstates of $\begin{cases} \text{Hamiltonian, eigenvalue } \frac{E_1}{n^2} \\ \vec{L}^2, \text{ eigenvalue } \hbar^2 l(l+1) \\ L_z, \text{ eigenvalue } \hbar m \end{cases}$

(a) $\langle E \rangle = \int d\vec{r} \left(\frac{4}{5}\psi_{100}^* - \frac{3i}{5}\psi_{211}^*\right) \hat{H} \left(\frac{4}{5}\psi_{100} + \frac{3i}{5}\psi_{211}\right) =$

$= \frac{16}{25} E_1 + \frac{9}{25} E_2 = \frac{16}{25} E_1 + \frac{9}{25}\frac{E_1}{4} =$

$= \frac{E_1}{25}\left(16 + \frac{9}{4}\right) = \frac{73}{100} E_1$

(b) $\langle \vec{L}^2 \rangle = \int d\vec{r} \left(\frac{4}{5}\psi_{100}^* - \frac{3i}{5}\psi_{211}^*\right) \vec{L}^2 \left(\frac{4}{5}\psi_{100} + \frac{3i}{5}\psi_{211}\right) =$

$= \frac{16}{25}\hbar^2 0(0+1) + \frac{9}{25}\hbar^2 1(1+1) = \frac{18}{25}\hbar^2$

$\langle L_z \rangle = \frac{16}{25}\hbar \cdot 0 + \frac{9}{25}\hbar \cdot 1 = \frac{9}{25}\hbar$

(c) $\psi(t) = \frac{4}{5}\psi_{100} e^{-iE_1 t/\hbar} + \frac{3i}{5}\psi_{211} e^{-iE_2 t/\hbar}$

None of the expectation values in (a, b) depend on time because $\int d\vec{r}\, \psi_{100}^* \psi_{211} = 0$ always.

P4 $\chi = A \begin{pmatrix} i \\ 3+i \end{pmatrix}$

(a) $1 = |A|^2 (-i, 3-i) \begin{pmatrix} i \\ 3+i \end{pmatrix} = |A|^2 \left(1 + (3-i)(3+i)\right) =$

$= 11\,|A|^2 \Rightarrow A = \frac{1}{\sqrt{11}}$

(b) $\chi = \frac{1}{\sqrt{11}} \begin{pmatrix} i \\ 3+i \end{pmatrix} = c_1 \chi_+^{(z)} + c_2 \chi_-^{(z)} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$c_1 = (1,0)\chi = (1,0) \frac{1}{\sqrt{11}} \begin{pmatrix} i \\ 3+i \end{pmatrix} = \frac{i}{\sqrt{11}}$

$c_2 = (0,1)\chi = (0,1) \frac{1}{\sqrt{11}} \begin{pmatrix} i \\ 3+i \end{pmatrix} = \frac{3+i}{\sqrt{11}}$

$S_z$ will give : $\begin{cases} +\frac{\hbar}{2} & \text{with } p_+ = \frac{1}{11} = |c_1|^2 \\ -\frac{\hbar}{2} & p_- = \left|\frac{3+i}{\sqrt{11}}\right|^2 = \frac{10}{11} \end{cases}$

$\langle S_z \rangle = \frac{\hbar}{2}\left(\frac{1}{11}\right) + \left(-\frac{\hbar}{2}\right)\frac{10}{11} = -\frac{\hbar}{2}\left(\frac{9}{11}\right)$

Check: $\langle S_z \rangle = \frac{1}{11}(-i, 3-i)\frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} i \\ 3+i \end{pmatrix} =$

$= \frac{1}{11}\frac{\hbar}{2}(-i, 3-i)\begin{pmatrix} i \\ -3-i \end{pmatrix} = \frac{1}{11}\frac{\hbar}{2}\left(1 - (3-i)(3+i)\right) =$

$= \frac{1}{11}\frac{\hbar}{2}\left(1 - (9+1)\right) = -\frac{9}{11}\frac{\hbar}{2}.$

(c) $\chi = c_1 \chi_+^{(x)} + c_2 \chi_-^{(x)} = c_1 \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$c_1 = \frac{1}{\sqrt{2}}(1,1)\frac{1}{\sqrt{11}}\begin{pmatrix} i \\ 3+i \end{pmatrix} = \frac{1}{\sqrt{22}}(i + 3 + i) = \frac{3+2i}{\sqrt{22}}$

$c_2 = \frac{1}{\sqrt{2}}(1,-1)\frac{1}{\sqrt{11}}\begin{pmatrix} i \\ 3+i \end{pmatrix} = \frac{1}{\sqrt{22}}(i - 3 - i) = -\frac{3}{\sqrt{22}}$

$S_x$ will give: $+\frac{\hbar}{2}$ with $p_+ = \left|\frac{3+2i}{\sqrt{22}}\right|^2 = \frac{9+4}{22} = \frac{13}{22}$

$-\frac{\hbar}{2}$ $\quad p_- = \frac{9}{22}$

$\langle S_x \rangle = \frac{\hbar}{2}\,\frac{13-9}{22} = \frac{\hbar}{11}$

(d) $\chi = c_1 \chi_+^{(y)} + c_2 \chi_-^{(y)} = c_1 \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} + c_2 \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}$

$$c_1 = \frac{1}{\sqrt{2}}(1,-i)\frac{1}{\sqrt{11}}\begin{pmatrix}i\\ 3+i\end{pmatrix} = \frac{1}{\sqrt{22}}(i-3i+1) = \frac{1-2i}{\sqrt{22}}$$

$$c_2 = \frac{1}{\sqrt{2}}(1,i)\frac{1}{\sqrt{11}}\begin{pmatrix}i\\ 3+i\end{pmatrix} = \frac{1}{\sqrt{22}}(i+3i-1) = \frac{-1+4i}{\sqrt{22}}$$

$S_y$ will give : $\begin{cases} +\frac{\hbar}{2} & \text{with } p_+ = \left|\frac{1-2i}{\sqrt{22}}\right|^2 = \frac{5}{22} \\ -\frac{\hbar}{2} & p_- = \left|\frac{-1+4i}{\sqrt{22}}\right|^2 = \frac{17}{22} \end{cases}$

$$\langle S_y\rangle = \frac{\hbar}{2}\left(\frac{5}{22}-\frac{17}{22}\right) = -\frac{12}{22}\frac{\hbar}{2} = -\frac{3}{11}\hbar$$

(e) Classically $\langle S_x\rangle = \sin\theta\cos\varphi$

$\langle S_y\rangle = \sin\theta\sin\varphi$

$\langle S_z\rangle = \cos\theta$

$$\Rightarrow \tan\varphi = \frac{\langle S_y\rangle}{\langle S_x\rangle} = \frac{-3\hbar/11}{\hbar/11} = -3$$

$$\tan\theta = \frac{\sqrt{\langle S_x\rangle^2 + \langle S_y\rangle^2}}{\langle S_z\rangle} = \frac{\sqrt{\left(\frac{1}{11}\right)^2 + \left(\frac{3}{11}\right)^2}}{-\frac{9}{22}} = \frac{\sqrt{10}}{11}\left(-\frac{22}{9}\right) = -\frac{2\sqrt{10}}{9}$$

$$\left.\begin{aligned} \varphi &= \arctan(-3) = -71.6^\circ \\ \theta &= \arctan\left(-\frac{2\sqrt{10}}{9}\right) = -35.1^\circ \end{aligned}\right\}$$

it is ok not to give these numerical values as they do require calculator.

$\boxed{P4}$

$$S_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \qquad S_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad S_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

(a)
$$[S_y, S_z] = \frac{\hbar^2}{\sqrt{2}} \left\{ \underbrace{\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}}_{\begin{pmatrix} 0 & 0 & 0 \\ i & 0 & i \\ 0 & 0 & 0 \end{pmatrix}} - \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}}_{\begin{pmatrix} 0 & -i & 0 \\ 0 & 0 & 0 \\ 0 & -i & 0 \end{pmatrix}} \right\}$$

$$= \frac{\hbar^2}{\sqrt{2}} \begin{pmatrix} 0 & i & 0 \\ i & 0 & i \\ 0 & i & 0 \end{pmatrix} = i\hbar \cdot \underbrace{\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}}_{S_x}$$

as it should be

(b)
$$S_x^2 + S_y^2 + S_z^2 = \hbar^2 \, 1(1+1) \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 2\hbar^2.$$

matrix unity.

Indeed,
$$S_x^2 = \frac{\hbar^2}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$S_y^2 = \frac{\hbar^2}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$S_z^2 = \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow S_x^2 + S_y^2 + S_z^2 = \hbar^2 \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} + \hbar^2 \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$+ \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \hbar^2 \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = 2\hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(c)
$$H = \frac{a}{\hbar^2} S_z^2 + \frac{b}{\hbar^2}\left(S_x^2 - S_y^2\right) = a\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 1\end{pmatrix} +$$

$$+ b\left\{\begin{pmatrix}\frac{1}{2} & 0 & \frac{1}{2}\\ 0 & 1 & 0\\ \frac{1}{2} & 0 & \frac{1}{2}\end{pmatrix} - \begin{pmatrix}\frac{1}{2} & 0 & -\frac{1}{2}\\ 0 & 1 & 0\\ -\frac{1}{2} & 0 & \frac{1}{2}\end{pmatrix}\right\} =$$

$$= a\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 1\end{pmatrix} + b\begin{pmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ 1 & 0 & 0\end{pmatrix} = \begin{pmatrix}a & 0 & b\\ 0 & 0 & 0\\ b & 0 & a\end{pmatrix}$$

Eigenvalues: $\det\left(H - \lambda \hat{1}\right) = 0$

$$\Rightarrow \det\begin{pmatrix}a-\lambda & 0 & b\\ 0 & -\lambda & 0\\ b & 0 & a-\lambda\end{pmatrix} = (a-\lambda)\begin{vmatrix}-\lambda & 0\\ 0 & a-\lambda\end{vmatrix} + b\begin{vmatrix}0 & -\lambda\\ b & 0\end{vmatrix} =$$

$$= -\lambda(a-\lambda)^2 + b^2\lambda = \lambda\left(b^2 - (a-\lambda)^2\right) = 0$$

$$\Rightarrow \lambda_1 = 0, \quad \pm b = a - \lambda \Rightarrow \lambda_{2,3} = a \pm b$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = a+b, \lambda_3 = a-b$$

(d) Eigenstates

$$\begin{pmatrix}a & 0 & b\\ 0 & 0 & 0\\ b & 0 & a\end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix} = \lambda\begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix} \Rightarrow \begin{cases} a\alpha_1 + b\alpha_3 = \lambda\alpha_1\\ 0 = \lambda\alpha_2\\ b\alpha_1 + a\alpha_3 = \lambda\alpha_3\end{cases}$$

$\underline{\lambda_1 = 0}$
$$\left.\begin{aligned} a\alpha_1 &= -b\alpha_3\\ b\alpha_1 &= -\alpha\alpha_3\end{aligned}\right\} \Rightarrow a\alpha_1 = +\frac{b^2}{\alpha}\alpha_1 \Rightarrow \begin{cases}\alpha_1 = 0\\ \alpha_3 = 0\end{cases}$$

$$\Rightarrow \psi_1 = \begin{pmatrix}0\\ \alpha_2\\ 0\end{pmatrix} \Rightarrow \text{normalization requires } \alpha_2 = 1 : \psi_1 = \begin{pmatrix}0\\ 1\\ 0\end{pmatrix}$$

$\underline{\lambda_2 = a+b}$
$$a\alpha_1 + b\alpha_3 = (a+b)\alpha_1 \Rightarrow \alpha_3 = \alpha_1$$

$$\psi_2 = \begin{pmatrix}\alpha_1\\ \overset{0=\alpha_2}{0}\\ \alpha_1\end{pmatrix} \Rightarrow \text{normalizing: } \psi_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 0\\ 1\end{pmatrix}.$$

$\underline{\lambda_3 = a-b}$ $\quad \left.\begin{aligned} a\alpha_1 + b\alpha_3 &= (a-b)\alpha_1 \\ 0 &= (a-b)\alpha_2 \end{aligned}\right\} \Rightarrow \begin{aligned} \alpha_3 &= -\alpha_1 \\ \alpha_2 &= 0 \end{aligned}$

$$\psi_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

(e) $\psi(0) = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \sum_{i=1}^{3} r_i \psi_i$

where $r_i = \langle \psi_i | \psi(0) \rangle$

$$r_1 = (0, 1, 0)\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = c_2$$

$$r_2 = \frac{1}{\sqrt{2}}(1, 0, 1)\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \frac{c_1 + c_3}{\sqrt{2}}$$

$$r_3 = \frac{1}{\sqrt{2}}(1, 0, -1)\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \frac{c_1 - c_3}{\sqrt{2}}$$

$$\psi(0) = c_2 \psi_1 + \frac{c_1 + c_3}{\sqrt{2}}\psi_2 + \frac{c_1 - c_3}{\sqrt{2}}\psi_3 =$$

$$= c_2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{c_1 + c_3}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{c_1 - c_3}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} =$$

$$= c_2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{c_1 + c_3}{2}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{c_1 - c_3}{2}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Thus $\psi(t) = \begin{pmatrix} 0 \\ c_2 \\ 0 \end{pmatrix} + \frac{c_1+c_3}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{-i(a+b)t/\hbar} +$

$$+ \frac{c_1-c_3}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} e^{-i(a-b)t/\hbar}$$

Can simplify a little bit more:

$$\frac{1}{2} e^{-iat/\hbar} \left\{ \begin{pmatrix} (c_1+c_3)\, e^{-ibt/\hbar} \\ 0 \\ (c_1+c_3)\, e^{-ibt/\hbar} \end{pmatrix} + \begin{pmatrix} (c_1-c_3)\, e^{ibt/\hbar} \\ 0 \\ -(c_1-c_3)\, e^{ibt/\hbar} \end{pmatrix} \right\} =$$

$$= \frac{1}{2} e^{-iat/\hbar} \begin{pmatrix} 2c_1 \cos(bt/\hbar) - 2i\, c_3 \sin(bt/\hbar) \\ 0 \\ -2i\, c_1 \sin(bt/\hbar) + 2\, c_3 \cos(bt/\hbar) \end{pmatrix}$$

$$\Rightarrow \psi(t) = \begin{pmatrix} e^{-iat/\hbar} \left[ c_1 \cos(bt/\hbar) - i\, c_3 \sin(bt/\hbar) \right] \\ c_2 \\ e^{-iat/\hbar} \left[ -i\, c_1 \sin(bt/\hbar) + c_3 \cos(bt/\hbar) \right] \end{pmatrix}$$

<u>Check:</u> $\psi(t) \xrightarrow[t \to 0]{} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$

Thus, for $\chi(0) = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ we obtain from the general expression above

$$\chi(t) = \frac{1}{\sqrt{3}} \begin{pmatrix} e^{-iat/\hbar}\, e^{-ibt/\hbar} \\ -1 \\ e^{-iat/\hbar}\, e^{-ibt/\hbar} \end{pmatrix}$$